$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$\int d\phi = \int [(y^2 \cos x + z^3) dx] + \int [(2y \sin x - 4) dy] + \int [(3xz^2 - 2) dz]$$

$$\boxed{\phi = y^2 \sin x + z^3 x + y^2 \sin x - 4y + xz^3 - 2x + c}$$

Work done:

6. In a moving ~~Particle in a force field~~ $\bar{F} = 3x^2 \bar{i} + (2xz - y)\bar{j} + z\bar{k}$ along the straight line $O(0,0,0)$ to $A(2,1,3)$

1(a)

Sol: Given that $\bar{F} = 3x^2 \bar{i} + (2xz - y)\bar{j} + z\bar{k}$ & straight line from $O(0,0,0)$ to $A(2,1,3)$

equation OA is $\frac{x-0}{2} = \frac{y-0}{1} = \frac{z-0}{3} = t$ (say)

$x = 2t \quad y = t \quad z = 3t$

$dx = 2dt \quad dy = dt \quad dz = 3dt$

$t \to 0$ to 1

$$\bar{F} \cdot dr = [3x^2 \bar{i} + (2xz - y)\bar{j} + z\bar{k}][dx\,\bar{i} + dy\,\bar{j} + dz\,\bar{k}]$$

$$3x^2 dx + (2xz - y)\, dy + z\, dz$$

work done $\int_C \bar{F} \cdot d\bar{r} = \int_{t=0}^{1} [3(4t^2)(2dt) + (12t^2 - t)dt + 3t(3dt)]$

$$= \int_{t=0}^{1} [3(4t^2)(2dt) + (12t^2 - t)dt + 3t(3dt)]$$

$$= \int_{t=0}^{1} (24t^2 + 12t^2 - t + 9t)\, dt$$

$$= \int_{t=0}^{1} (36t^2 + 8t)\, dt = \left[\frac{36t^3}{3} + \frac{8t^2}{2}\right]_0^1$$

$$= \frac{\overset{12}{\cancel{36}}}{\cancel{3}} + \frac{\overset{4}{\cancel{8}}}{\cancel{2}}$$

$$= 12 + 4 = 16 \text{ units}$$

1/6/04

7. Find the work done by the force $\bar{F} = (2x - y - z)\bar{i} + (x + y + z)\bar{j} + (3x - 2y - 5z)\bar{k}$ along the curve C in xy-plane is given by $x^2 + y^2 = 9$ & $z = 0$.

1(b)

Sol: Given that $\bar{F} = (2x - y - z)\bar{i} + (x + y + z)\bar{j} + (3x - 2y - 5z)\bar{k}$ and also $x^2 + y^2 = 9$ & $z = 0$.

$V = ZI$ & $I = YV$

$$\bar{F} = (2x-y)\bar{i} + (x+y)\bar{j} + (3x-2y)\bar{k}$$

$$d\bar{r} = dx\,\bar{i} + dy\,\bar{j}$$

$$\bar{F}\cdot d\bar{r} = (2x-y)\,dx + (x+y)\,dy$$

take $x = 3\cos\theta$ $\qquad y = 3\sin\theta$

$dx = -3\sin\theta\, d\theta$ $\qquad dy = 3\cos\theta\, d\theta$

$\theta \to 0$ to 2π

$$\text{Work done} = \int_C \bar{F}\cdot d\bar{r}$$

$$= \int_{\theta=0}^{2\pi} (6\cos\theta - 3\sin\theta)(-3\sin\theta\, d\theta) + (3\cos\theta + 3\sin\theta)3\cos\theta\, d\theta$$

$$= \int_{\theta=0}^{2\pi} [-18\sin\theta\cos\theta + 9\sin^2\theta + 9\cos^2\theta + 9\sin\theta\cos\theta]\, d\theta$$

$$= \int_{\theta=0}^{2\pi} [9 - 9\sin\theta\cos\theta)\, d\theta$$

$$= \frac{9}{2}\int_{\theta=0}^{2\pi} [2 - 2\sin\theta\cos\theta]\, d\theta$$

$$= \frac{9}{2}\int_{\theta=0}^{2\pi} (2 - \sin 2\theta)\, d\theta$$

$$= \frac{9}{2}\int_{\theta=0}^{2\pi} (2 - \sin 2\theta)\, d\theta = \frac{9}{2}\left[2\theta + \frac{\cos 2\theta}{2}\right]_0^{2\pi}$$

$$= \frac{9}{2}\left[4\pi + \frac{\cos 4\pi}{2} - 0 - \frac{1}{2}\right]$$

$(\cos 4\pi = 1)$

$$= \frac{9}{2}(4\pi) = 18\pi$$

$dy = 3x^2\,dx$

$x \to 1$ to 2 from $(1,1)$ to $(2,8)$

$$\int_C \bar{F}\cdot d\bar{r} = \int_1^2 (5x^4 - 6x^2)\,dx + (2x^3 - 4x)\,3x^2\,dx$$

$$= \int_1^2 (5x^4 - 6x^2 + 6x^5 - 12x^3)\,dx$$

$$= \left[\frac{20x^5}{5} - 12x + \frac{30x^6}{6} - 36\frac{x^2}{2}\right]$$

$$= \left[\frac{5x^5}{5} - \frac{6x^3}{3} + \frac{6x^6}{6} - 12\frac{x^4}{4}\right]_1^2$$

$$= \left[x^5 - 2x^3 + x^6 - 3x^4\right]_1^2$$

$$= \left[(2)^5 - 2(2)^3 + (2)^6 - 3(2)^4\right] - \left[1 - 2 + 1 - 3\right]$$

$$= \left[32 - 16 + 64 - 48\right] - \left[-3\right]$$

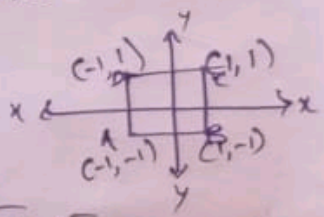
$$= 96 - 64 + 3$$

$$= 99 - 64 = 35.$$

06/06/24

3. Evaluate the line integral $\int_C (x^2+xy)\,dx + (x^2+y^2)\,dy$ where C is the square formed by the lines $x = \pm 1$ and $y = \pm 1$.

Sol: Given that $\int_C (x^2+xy)\,dx + (x^2+y^2)\,dy$ and the lines are



$x = \pm 1$ & $y = \pm 1$

$$\int_C \bar{F}\cdot d\bar{r} = \int_{AB} \bar{F}\cdot d\bar{r} + \int_{BC} \bar{F}\cdot d\bar{r} + \int_{CD} \bar{F}\cdot d\bar{r} + \int_{DA} \bar{F}\cdot d\bar{r} \to (1)$$

Along AB: $y = -1 \Rightarrow dy = 0$

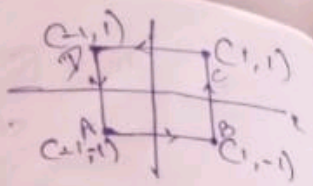
$x = -1$ to 1

$$\int (x^2 + xy)\,dx + \int (x^2+y^2)\,dy = \int_{x=-1}^{1} (x^2 - x)\,dx + 0$$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2}\right]_{-1}^{1}$$

$$= \frac{+1}{3} - \cancel{\frac{1}{2}} + \frac{1}{3} + \cancel{\frac{1}{2}} = +\frac{2}{3} \to (1)$$

Along BC :



$x = 1 \Rightarrow dx = 0$

$y = -1$ to 1

$$\int (x^2+xy)dx + (x^2+y^2)dy = \int_{y=-1}^{1} 0 + (1+y^2)\,dy$$

$$= \int_{-1}^{1} (1+y^2)\,dy$$

$$= \left[y + \frac{y^3}{3}\right]_{-1}^{1}$$

$$= 1 + \frac{1}{3} + 1 + \frac{1}{3} = \frac{8}{3} \rightarrow (2)$$

$$= \frac{8}{3}$$

Along CD :

$y = 1 \Rightarrow dy = 0$

$x \rightarrow 1$ to -1

$$\int (x^2+xy)dx + (x^2+y^2)dx = \int_{1}^{-1} (x^2+x)\,dx$$

$$= \left[\frac{x^3}{3} + \frac{x^2}{2}\right]_{1}^{-1}$$

$$= \left(-\frac{1}{3} - \frac{1}{2} - \frac{1}{3} + \frac{1}{2}\right) = -\frac{1}{3} + \frac{1}{2} - \frac{1}{3} - \frac{1}{2}$$

$$= -\frac{2}{3} = 0 \rightarrow (3) \qquad = -\frac{2}{3} \rightarrow (3)$$

Along DA :

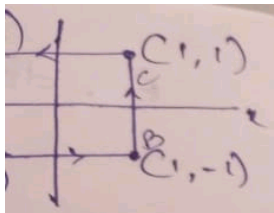
$x = 1 \Rightarrow dx = 0$

$y \rightarrow 1$ to -1

$$\int (x^2+xy)\,dx + (x^2+y^2)\,dy = \int_{1}^{-1} (1+y^2)\,dy$$

$$= \left[y + \frac{y^3}{3}\right]_{1}^{-1}$$

$$= -1 - \frac{1}{3} - 1 - \frac{1}{3} = -\frac{8}{3} \rightarrow (4)$$



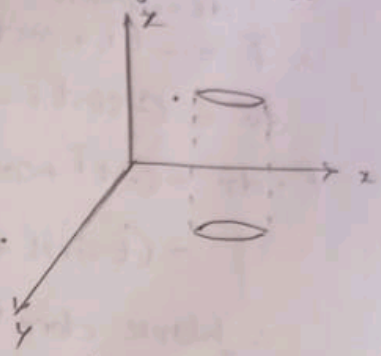
$$\int_C (x^2+xy)\,dx + (x^2+y^2)\,dy = \left(\frac{0}{3}+0+0+0\right)$$
$$= 0.$$

$$= \frac{2}{3}+\frac{8}{3}-\frac{2}{3}-\frac{8}{3}$$
$$= 0 \;//$$

Compute the line integral $\int (y^2\,dx - x^2\,dy)$ around the

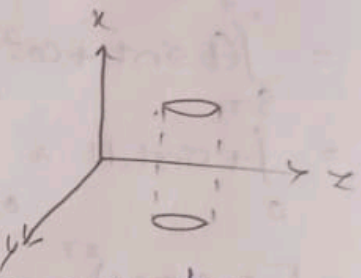
(or) xz-plane.

Case-1: Let 'R' be the orthogonal projection of 's' on the xy-plane $ds = \frac{dx\,dy}{|\hat{n}\cdot\bar{k}|}$



$$\boxed{\iint_S \bar{F}\cdot\hat{n}\,ds = \iint_S \bar{F}\cdot\hat{n}\cdot\frac{dx\,dy}{|\hat{n}\cdot\bar{k}|}}$$

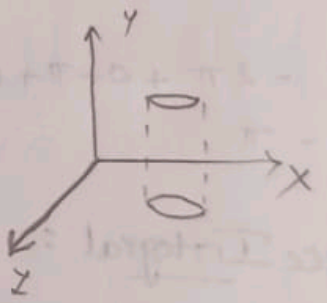
Case-2:
In the surface 's' is projected on yz-plane $ds = \frac{dx\,dy}{|\hat{n}\cdot\bar{i}|}$



$$\boxed{\iint_S \bar{F}\cdot\hat{n}\,ds = \iint_S \bar{F}\cdot\hat{n}\cdot\frac{dx\,dy}{|\hat{n}\cdot\bar{i}|}}$$

Case-3:
In the surface 's' is projected on zx-plane

$ds = \frac{dx\,dy}{|\hat{n}\cdot\bar{j}|}$



$$\boxed{\iint_S \bar{F}\cdot\hat{n}\,ds = \iint_S \bar{F}\cdot\hat{n}\cdot\frac{dx\,dy}{|\hat{n}\cdot\bar{j}|}}$$

If $\bar{F} = 18z\bar{i} - 12\bar{j} + 3y\bar{k}$ evaluate $\int_S \bar{F}\cdot\bar{n}\,ds$ where 's' is the surface of the plane ~~2x+3y~~+6z=12 in the first octant

Sol

Given $\bar{F} = 18z\bar{i} - 12\bar{j} + 3y\bar{k}$

Where 'R' be the projection of 's' on the xy-plane

$$\int_S \bar{F}\cdot\hat{n}\,ds = \iint_S \bar{F}\cdot\hat{n}\,\frac{dx\,dy}{|\hat{n}\cdot\bar{k}|} \longrightarrow (1)$$

Let $\phi = 2x + 3y + 6z - 12$

then $\nabla\phi = \left[\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right](2x+3y+6z-12)$

$$\nabla\phi = 2\bar{i} + 3\bar{j} + 6\bar{k}$$

$$|\nabla\phi| = \sqrt{4+9+36} = \sqrt{49} = 7$$

WKT, $\hat{n} = \dfrac{\nabla\phi}{|\nabla\phi|} = \dfrac{2\bar{i}+3\bar{j}+6\bar{k}}{7}$

$$\bar{F}\cdot\hat{n} = (18x\bar{i} - 12\bar{j} + 3y\bar{k})\,\frac{2\bar{i}+3\bar{j}+6\bar{k}}{7}$$

$$= \frac{36x - 36 + 18y}{7}$$

$$= \frac{6}{7}(6x - 6 + 3y)$$

$$\hat{n}\cdot\bar{k} = \left(\frac{2\bar{i}+3\bar{j}+6\bar{k}}{7}\right)(\bar{k}) = \frac{6}{7}$$

$\left(\begin{array}{l}\bar{i}\cdot\bar{k}=0\\ \bar{j}\cdot\bar{k}=0\\ \bar{k}\cdot\bar{k}=1\end{array}\right)$

But, $2x+3y+6z=12$

$$z = \frac{1}{6}(12-2x-3y)$$

$$z=0 \Rightarrow 2x+3y=12$$

$$y = \frac{12-2x}{3}$$

$y \to 0$ to $\frac{12-2x}{3}$

$$y=0 \Rightarrow 12-2x=0$$

$$x=6$$

$x \to 0$ to 6

$$\int_S \bar{F}\cdot\hat{n}\,ds = \iint_S \bar{F}\cdot\hat{n}\,\frac{dx\,dy}{|\hat{n}\cdot\bar{k}|}$$

$$= \int_{x=0}^{6}\int_{y=0}^{\frac{12-2x}{3}} \frac{6}{7}\left\{6\left(\frac{1}{6}(12-2x-3y)\right) - 6 + 3y\right\}\frac{7}{6}\,dx\,dy$$

$$= \int_{x=0}^{6}\left[\int_{y=0}^{\frac{12-2x}{3}} (6-2x)\,dy\right]dx$$

$$= \int_{x=0}^{6}(6-2x)\,[y]_0^{\frac{12-2x}{3}}\,dx = \int_0^6 2(3-x)\left(\frac{12-2x}{3}\right)dx$$

$$= \frac{4}{3}\int_0^6 (3-x)(6-x)\,dx$$

$$= \frac{4}{3}\int_0^6 (18-9x+x^2)\,dx$$

$$= \frac{4}{3}\left[18x - \frac{9x^2}{2} + \frac{x^3}{3}\right]_0^6 = \frac{4}{3}[108-162+72]$$

$$= \frac{4}{3}\times 18$$

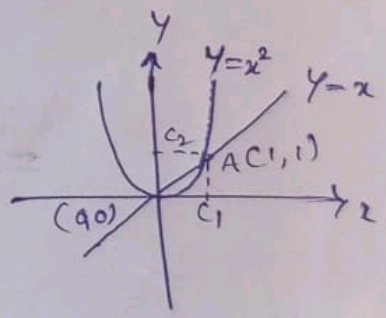
$$= \frac{72}{3} = 24$$

3. Verify Green's theorem for $\oint_C (xy+y^2)\,dx + x^2\,dy$ where 'c' is bounded by $y=x$ & $y=x^2$.

Sol Given that

$$\oint_C (xy+y^2)\,dx + x^2\,dy$$

and also, $y=x$ and $y=x^2$



By green's theorem we have,

$$\oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

Here $M = xy+y^2$ $\qquad N = x^2$

$$\frac{\partial M}{\partial y} = x+2y \qquad \frac{\partial N}{\partial x} = 2x$$

RHS Here $x \to 0$ to 1

$y \to x^2$ to x,

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy = \int_{x=0}^{1} \int_{y=x^2}^{x} (2x - x - 2y)\,dy\,dx$$

$$= \int_{x=0}^{1} \left[\int_{y=x^2}^{x} (x-2y)\,dy \right] dx$$

$$= \int_{x=0}^{1} \left[xy - \frac{\cancel{2}y^2}{\cancel{2}} \right]_{x^2}^{x} dx$$

$$= \int_{x=0}^{1} (x^2 - x^2 - x^3 + x^4)\,dx$$

$$= \int_{x=0}^{1} (x^4 - x^3)\,dx$$

$$= \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20} \rightarrow (1)$$

<u>LHS</u>

$$\oint_C M\,dx + N\,dy = \oint_{C_1} (M\,dx + N\,dy) + \oint_{C_2} (M\,dx + N\,dy) \rightarrow (2)$$

<u>Along the curve C_1</u> : $y = x^2$

$dy = 2x\,dx$

$x \rightarrow 0$ to 1

$$\oint_C M\,dx + N\,dy \Rightarrow \oint_C (xy + y^2)\,dx + x^2\,dy = \int_{x=0}^{1} (x^3 + x^4 + 2x^3)\,dx$$

$$= \int_{x=0}^{1} (3x^3 + x^4)\,dx = \left[\frac{3x^4}{4} + \frac{x^5}{5} \right]_0^1$$

$$= \frac{3}{4} + \frac{1}{5} = \frac{19}{20} \rightarrow (3)$$

Along the curve C_2 :

$y = x$

$dx = dy \Rightarrow x \rightarrow 1$ to 0

$$\oint_{C_2} (xy + y^2)\,dx + x^2\,dy = \int_{x=1}^{0} (x^2 + x^2 + x^2)\,dx$$

$$= \int_{x=1}^{0} 3x^2\,dx = \left(\frac{3x^3}{3} \right)_1^0 = -1 \rightarrow (4)$$

Sub eq (3), (4) in eq (2)

$$\oint_C (xy + y^2)\,dx + x^2\,dy = \frac{19}{20} - 1 = \frac{-1}{20} \rightarrow (5)$$

From eq ① & eq ②

LHS = RHS

Hence, Green's theorem is proved.

Ex. $\oint (3x^2 - 8y^2)\,dx + (4y - 6x$

2. Verify Green's theorem in a plane for $\oint_C (3x^2-8y^2)dx+(4y-6xy)dy$ where 'c' is the region bounded by $y=\sqrt{x}$ & $y=x^2$

Sol: Given that $\oint_C (3x^2-8y^2)\,dx+(4y-6xy)\,dy$

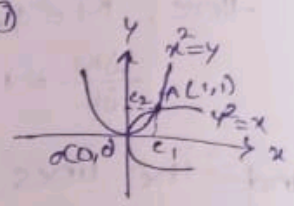
This is of the form $\oint_C M\,dx+N\,dy$

$M = 3x^2-8y^2$ $\qquad$ $N = 4y-6xy$

$\frac{\partial M}{\partial y} = -16y$ $\qquad$ $\frac{\partial N}{\partial x} = -6y$

From the Green's theorem

$$\oint_C M\,dx+N\,dy = \iint_R \left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)dx\,dy \rightarrow (1)$$



and also given that $y=\sqrt{x}$ & $y=x^2$

Along the curve C_1

$y=x^2$

$dy=2x\,dx$

$x \rightarrow 0$ to 1

$$\oint_C (3x^2-8y^2)dx+(4y-6xy)dy = \int_{x=0}^{1} (3x^2-8x^4)dx+(4x^2-6x^3)2x\,dx$$

$$= \int_{x=0}^{1} (3x^2-8x^4+8x^3-12x^4)\,dx$$

$$= \int_{x=0}^{1} (3x^2-20x^4+8x^3)\,dx$$

$$= \left[\frac{3x^3}{3}-\frac{20x^5}{5}+\frac{8x^4}{4}\right]_0^1$$

$$= 1-4+2 = -1. \rightarrow (2)$$

Along the curve C_2

$y=\sqrt{x}$

$dy = \frac{1}{2\sqrt{x}}dx$

$x \rightarrow 1$ to 0

$$\oint_C (3x^2-8y^2)dx+(4y-6xy)dy = \int_{x=1}^{0} (3x^2-8x)\,dx+(4\sqrt{x}-6x^{3/2})\frac{1}{2\sqrt{x}}$$

$$= \int_1^0 \left(3x^2-8x+\frac{4\sqrt{x}}{2\sqrt{x}}-\frac{6\sqrt{x}\cdot x}{2\sqrt{x}}\right)dx$$

$$= \int_1^0 (3x^2-8x+2-3x)\,dx = \int_1^0 (3x^2-11x+2)\,dx$$

ameters in terms of a

$$= \left[\frac{3x^3}{3} - \frac{11x^2}{2} + 2x\right]_1^0$$

$$= -1 + \frac{11}{2} - 2 = \frac{11}{2} - 3 = \frac{5}{2} \rightarrow (3)$$

$$\oint_C (3x^2 - 8y^2)\,dx + (4y - 6xy)\,dy = -1 + \frac{5}{2} \text{ (from eq (2) \& (3))}$$

$$= \frac{3}{2} \rightarrow (4)$$

RHS Here $x \to 0$ to 1
$y \to x^2$ to $\sqrt{x}$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy = \int_{x=0}^{1} \int_{y=x^2}^{\sqrt{x}} (-6y + 16y)\,dy\,dx$$

$$= 10 \int_{x=0}^{1} \left[\frac{y^2}{2}\right]_{x^2}^{\sqrt{x}} dx = 5 \int_{x=0}^{1} (x - x^4)\,dx$$

$$= 5\left[\frac{x^2}{2} - \frac{x^5}{5}\right]_0^1 = 5\left(\frac{1}{2} - \frac{1}{5}\right)$$

$$= 5\left(\frac{3}{10}\right) = \frac{3}{2} \rightarrow (5)$$

From eq (4) & (5)

LHS = RHS

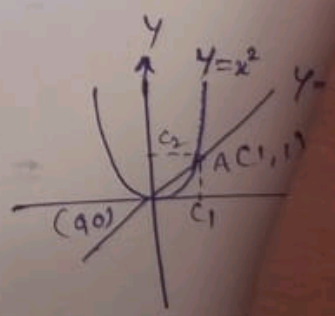
Hence, verified Green's theorem.

3. Verify Green's theorem for $\oint_C (xy + y^2)\,dx + x^2\,dy$ where 'C' is bounded by $y = x$ & $y = x^2$.

Sol Given that

$$\oint_C (xy + y^2)\,dx + x^2\,dy$$

and also, $y = x$ and $y = x^2$



By green's theorem we have,

$$\oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

Here $M = xy + y^2$ $\quad N = x^2$

$$\frac{\partial M}{\partial y} = x + 2y \qquad \frac{\partial N}{}$$

Where 'C' is traversed in the positive direc...

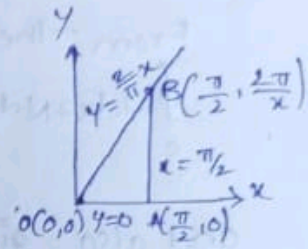
Note:

$$\iint_R dx\,dy = \text{area of the region.}$$

1. Evaluate by Green's theorem $\oint_C (y-\sin x)\,dx + \cos x\,dy$ where C is the triangle enclosed by the lines $y=0,\ x=\pi/2,\ \pi y = 2x$.

Sol Given that $\oint_C (y-\sin x)\,dx + \cos x\,dy$

This is of the form $\oint_C M\,dx + N\,dy$



Here, $M = y-\sin x$ $\quad N = \cos x$

$\frac{\partial M}{\partial y} = 1$ $\quad \frac{\partial N}{\partial x} = -\sin x$

and the lines are $y=0,\ x=\pi/2,\ \pi y = 2x$.

$y = \frac{2}{\pi}x$.

Here, $x \to 0$ to $\frac{\pi}{2}$

$y \to 0$ to $\frac{2x}{\pi}$

$$\oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

$$\oint_C (y-\sin x)\,dx + \cos x\,dy = \int_{x=0}^{\pi/2}\int_{y=0}^{\frac{2x}{\pi}} (-\sin x - 1)\,dy\,dx$$

$$= \int_{x=0}^{\pi/2} [-\sin x - 1]\,[y]_0^{\frac{2x}{\pi}}$$

$$= -\frac{2}{\pi}\int_{x=0}^{\pi/2} [x\sin x + x]\,dx$$

$$= -\frac{2}{\pi}\left[\left\{[x(-\cos x)]_0^{\pi/2} - \int_0^{\pi/2} 1(-\cos x)\,dx\right\} + \left[\frac{x^2}{2}\right]_0^{\pi/2}\right]$$

$$= -\frac{2}{\pi}\left[0 + 0 + [\sin x]_0^{\pi/2} + \frac{\pi^2}{8}\right]$$

$$= -\frac{2}{\pi}\left[1 + \frac{\pi^2}{8}\right]$$

$$= -\frac{2}{\pi}\left(1 + \frac{\pi^2}{8}\right) = -\frac{2}{\pi} - \frac{\pi}{4} = -\left(\frac{2}{\pi} + \frac{\pi}{4}\right)$$

$$\oint_C (y-\sin x)\,dx + \cos x\,dy = -\left(\frac{2}{\pi} + \frac{\pi}{4}\right)$$